Rigid Identity as an Inverse Limit in Observable Channels

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Abstract

We introduce a formal framework for rigid identity as a mathematical object emerging from the dynamics of observable systems under minimal structural constraints. Identity is defined as an inverse limit over a family of locally corrupted states linked by retroinductive projection maps, yielding a globally invariant object despite persistent environmental perturbations.

For a common metric tower, exact fiber lifts with bounded weighted amplification yield a Hausdorff bound. Commuting coordinate homeomorphisms induce an isomorphism, realized explicitly by a ramified circle tower with positive deformation. The deformation-resistance ratio, historically termed ontological mass, depends on the declared intervention cost and comparison metric.

For exact comparisons of model histories, one finite-cost intervention with positive deformation certifies a finite resistance ratio. Estimating that ratio from observable data requires a calibrated comparison and coverage of the intervention class.

Using these results, we demonstrate a strong uniqueness theorem: within a single observable channel, no two distinct rigid identities can coexist. Identity is therefore neither role-based nor transferable; it is structurally indivisible once instantiated.

Finite binary representations face a fidelity bound when the specified responses are sufficiently separated. A latent dimension or an intervention cost alone does not determine the representation budget.

This work reframes identity as a categorical invariant of system dynamics, independent of narrative descriptions or biological assumptions, and establishes a foundation for distinguishing structurally real identities from simulable behavioral approximations.

1 Scope, System Boundary, and Non-Inference Note

Scope and admissible reading. This paper defines rigid identity as an inverse-limit object over observable state channels, derives rigidity and deformation-rigidity structure under explicit projective, persistence, and perturbation hypotheses, proves channel-level uniqueness, and states an operational detectability criterion inside that model. The historical phrase "ontological mass" is used as a technical alias for the scalar rigidity/deformation modulus M_Ω, not as a claim of physical ontology. The paper does not establish phenomenology, subjective experience, biological realization, substrate-independent physical realization, or metaphysical priority over competing frameworks. Related systems may implement diagnostics, perturbation protocols, and evidence-governed comparisons motivated by the framework, but those outputs do not implement the inverse-limit identity object as a closed empirical fact and do not by themselves certify CCR membership for any present architecture. Behavioral indistinguishability, runtime survival under perturbation, and internal support artifacts therefore remain model-facing evidence surfaces rather than proof of rigid identity, consciousness, or claim closure.

2 Introduction

The concept of identity in dynamical systems has been studied from multiple angles: cognitive science treats it as emergent from neural processes, philosophy of mind debates its ontological

status, and computer science models it as state persistence. Yet a formal, substrate-independent characterization of identity as a *structural invariant* remains underdeveloped.

This work addresses a specific question: **Under what minimal conditions does a system exhibit an identity that is structurally rigid**—that is, invariant under perturbation, non-transferable, and externally detectable?

We study structural properties that emerge from well-defined mathematical constraints. All results presented here are conditional on the stated assumptions; they do not extend to systems that violate those assumptions.

2.1 Motivation

The motivation for this work arises from a simple observation: most existing theories of identity (in cognitive science, AI, and philosophy) define identity as an *emergent result* of underlying dynamics. This creates a fundamental vulnerability: if identity is emergent, it can be destabilized, transferred, or duplicated by modifying the dynamics.

We propose an alternative: identity as a *prior constraint* on dynamics, not a result of them. Under this view, the dynamics of a system occur *within* the identity, rather than producing it. This inversion has consequences:

1. Identity becomes non-transferable by construction.
2. Rigidity emerges as a structural necessity, not a contingent property.
3. External detectability becomes a well-posed problem.

2.2 Ontological Closure

Appendix A establishes that, under the stated persistence and observability constraints, the central structures of this framework—the inverse-limit identity, the $\aleph$ operator, causal closure, and retroinductive structure—are model-relative necessities rather than arbitrary design choices. The framework does not describe an unconstrained "system"; it describes the class selected by those hypotheses.

3 Minimal Assumptions and Framework

We begin by stating the minimal assumptions required for the subsequent results. These assumptions are deliberately weak—any system that fails to satisfy them is not a candidate for rigid identity, and no claim is made about such systems.

3.1 Assumption A1: Observable Channel

There exists an observable channel $\mathcal{C} : \mathcal{U} \to \mathcal{O}$ mapping internal states $\mathcal{U}$ to observable outputs $\mathcal{O}$. We do not assume injectivity of $\mathcal{C}$; multiple internal states may produce identical observations.

3.2 Assumption A2: Temporal Persistence

The system persists through discrete time steps $t \in \mathbb{N}$. At each step, the internal state I_t is subject to:

- Environmental input E_t (not fully controlled by the system)
- Internal dynamics governed by transition function F

3.3 Assumption A3: Non-Triviality (R0)

The system satisfies a minimal non-triviality constraint:

1. **Persistence:** $\forall t, I_t \neq \varnothing$
2. **Non-reactivity:** $\nexists f : \mathcal{E} \to \mathcal{I}$ such that $I_{t+1} = f(E_t)$
3. **Non-constancy:** $\nexists c$ such that $\forall t, I_t = c$

This prohibits only three trivial cases: dead systems, pure mirrors, and frozen states. Any system failing R0 is not a candidate for identity analysis.

3.4 Categorical Regularity: Topological Structure of State Spaces

To guarantee existence and uniqueness of inverse limits, and to enable rigorous stability analysis, we impose the following structural condition:

Definition 3.1 (Observable State Spaces)**.** All state spaces S_t are objects of a concrete category $\mathbf{C}$ whose:

- **Objects** are compact Hausdorff topological spaces;
- **Morphisms** are continuous surjective maps.

Axiom 3.2 (State Space Regularity)**.** Each state space S_t in the projective system is a compact metric space with compatible metric d_t. Consequently, each S_t is compact Hausdorff as a topological space and Polish as a metric space. The inverse limit $\mathcal{I}$ inherits the corresponding compact metrizable subspace structure from the product.

Remark 3.3 (Exclusion of non-compact Polish spaces). Axiom 3.2 refines Definition 3.1 rather than weakening it. Non-compact Polish spaces such as $\mathbb{R}^n$ are not covered by this compact inverse-limit argument. Any extension of the CCR framework to non-compact settings requires additional tightness, compactification, or coercivity hypotheses and is not automatic.

Hypothesis 3.4 (Topological Regularity)**.** All projections $\pi_{t+1 \to t} : S_{t+1} \to S_t$ are continuous surjections between compact Hausdorff spaces.

Remark 3.5. This condition is sufficient for the compact inverse-limit construction used here. By Tychonoff's theorem, the product $\prod_t S_t$ is compact when each S_t is compact. Furthermore, the inverse limit $\varprojlim S_t$ is a closed subspace of the product, hence compact. This guarantees:

1. Existence of the inverse limit (non-emptiness under surjectivity);
2. Well-defined metrics and convergence;
3. Applicability of functional analysis tools.

Proposition 3.6 (A0 Becomes Categorially Closed)**.** *Under Definition 3.1, Assumption A3 (R0) is not merely a constraint but defines a closed class of systems admitting rigorous analysis.*

3.5 Conditional Nature of Results

All theorems in this paper are conditional:

> *"If a system satisfies A1–A3, then the following properties hold..."*

We make no claims about systems that violate these assumptions. This conditional structure is essential: it means our results cannot be falsified by pointing to systems outside the assumption set, and cannot be extended beyond their stated scope without additional justification.

4 Inverse Limit Identity

4.1 Preliminaries and Notation

Let $\{S_t\}_{t\in\mathbb{N}}$ be a family of non-empty state spaces indexed by discrete time. Each S_t represents the admissible internal configurations of a system at time t.

We assume the existence of projection morphisms

$$\pi_{t+1\to t} : S_{t+1} \to S_t \tag{1}$$

satisfying the standard compatibility condition:

$$\pi_{t+1\to t} \circ \pi_{t+2\to t+1} = \pi_{t+2\to t} \quad \forall t \in \mathbb{N}. \tag{2}$$

No assumption is made regarding:

- the cardinality of S_t,
- continuity or smoothness,
- realizability by a computational mechanism.

The structure $(\{S_t\}, \{\pi_{t+1\to t}\})$ therefore defines a **projective (inverse) system** in the categorical sense [4, 1].

4.2 Definition of Identity as an Inverse Limit

Definition 4.1 (Identity as Inverse Limit)**.** We define identity as the inverse limit of the projective system $(S_t, \pi_{t+1\to t})$:

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_0, x_1, x_2, \ldots) \in \prod_{t=0}^{\infty} S_t \;\middle|\; \pi_{t+1\to t}(x_{t+1}) = x_t \;\; \forall t \right\}. \tag{3}$$

An element of $\mathcal{I}$ is thus a *globally consistent history*, not reducible to any finite prefix.

This definition is purely structural and does not depend on:

- observability,
- external inputs,
- internal dynamics.

4.3 Compatibility with Minimal Assumption R0

We now show that this construction is determined within the minimal restriction R_0.

Proposition 4.2 (Necessity of Projective Structure)**.** *Any system satisfying R_0 admits a projective representation $(S_t, \pi_{t+1\to t})$ whose inverse limit is non-empty.*

Proof. 1. Persistence implies that admissible states cannot vanish over time.

2. Non-reactivity forbids forward-determined transitions $S_t \to S_{t+1}$ driven by environment.

3. Therefore, admissibility at time t must be determined by compatibility with future states, not past ones.

4. Hence, the only admissible morphisms preserving identity under R_0 are projections from future to past, i.e., $\pi_{t+1\to t}$.

5. The compatibility condition ensures persistence of consistency. □

4.4 Identity Is Not a State

A critical consequence of the inverse limit construction is that identity is not an element of any S_t.

Proposition 4.3 (Non-locality of Identity)**.** *For any t, there exists no injective map $i_t : \mathcal{I} \hookrightarrow S_t$.*

Proof. Suppose such an injective embedding existed. Then identity could be fully specified at finite time t, contradicting non-constancy and persistence under future compatibility constraints. Therefore, identity cannot be localized to any finite temporal slice. □

This formalizes a key distinction:

- S_t: local, time-indexed configurations
- $\mathcal{I}$: global, trans-temporal constraint

4.5 Impossibility of Progressive Construction

We now establish that identity cannot be constructed via progressive (forward) dynamics.

Theorem 4.4 (No Progressive Identity Construction)**.** *There exists no sequence of functions $F_t : S_t \to S_{t+1}$ such that*

$$\mathcal{I} = \lim_{t\to\infty} F_t \circ \cdots \circ F_0(S_0). \tag{4}$$

Proof. Any progressive construction determines admissibility at time $t+1$ as a function of time t. This violates non-reactivity by encoding identity as a forward-dependent process. Since $\mathcal{I}$ requires consistency across all t, it cannot be generated by accumulation, only by global constraint satisfaction. □

4.6 Minimality of the Inverse Limit Construction

Theorem 4.5 (Minimal Representation)**.** *Among all structures satisfying R_0, the inverse limit construction introduces the minimal additional structure necessary to define non-trivial identity.*

Justification.

- No topology is assumed.
- No metric is assumed.
- No dynamics is assumed.
- Only compatibility constraints are imposed.

Any weaker structure fails to ensure persistence. Any stronger structure introduces unnecessary assumptions.

4.7 Summary of Section 3

1. Identity is rigorously defined as an inverse limit $\mathcal{I} = \varprojlim S_t$.
2. This construction is determined within minimal assumption R_0.
3. Identity is not reducible to any finite-time state.
4. Progressive or reactive architectures cannot implement this structure.
5. No metaphysical assumptions are introduced.

This completes the formal construction of identity used throughout the remainder of the paper.

4.8 Identity Neutrality Clause

Throughout this paper, the identity object $\mathcal{I}$ is treated as a *purely abstract topological and categorical structure*. No psychological, biological, phenomenological, or experiential interpretation is assumed or required for any of the formal results derived herein.

The term "identity" refers exclusively to the role of $\mathcal{I}$ as an inverse-limit object encoding global temporal compatibility, and does not presuppose the existence of consciousness, subjective experience, selfhood, or any particular semantic content associated with identity in ordinary language.

All theorems, constructions, and stability results presented in this work are therefore statements about abstract mathematical objects only. Any interpretation of $\mathcal{I}$ in relation to psychological, phenomenological, or experiential concepts lies strictly outside the formal scope of this paper.

5 Rigidity Under Perturbations

5.1 Setup and Hypotheses

Let the projective system $(S_t, \pi_{t+1\to t})_{t\in\mathbb{N}}$ and its inverse limit $\mathcal{I} = \varprojlim S_t$ be as in Section 3. We introduce an external perturbation model and define deformed projections; then we state the technical hypotheses under which rigidity holds.

Hypothesis 5.1 (H1: Surjectivity / Lifting)**.** For every t, the projection $\pi_{t+1\to t} : S_{t+1} \to S_t$ is surjective.

This is a standard regularity condition for non-empty inverse limits; when necessary this can be replaced by a Mittag–Leffler condition or an explicit choice of right inverses.

Hypothesis 5.2 (H2: Compact Polish Metric Structure)**.** For each t, the state space S_t carries the compact metric d_t required by Axiom 3.2; in particular, (S_t, d_t) is Polish. All projection maps $\pi_{t+1\to t}$ are continuous with respect to d_t. The quantitative lifting estimate is a separate condition in Lemma 5.10.

Remark 5.3. The Polish structure ensures Borel regularity of all deformation families, enabling measurable selection theorems, probabilistic perturbation models, and the application of stochastic stability results to the ontological mass M_Ω. This is the standard regularity assumption in modern probability theory, ergodic theory, and descriptive set theory.

These assumptions are technical; they do not change the conceptual content (identity as inverse limit) but permit quantitative stability estimates and connect the framework to probabilistic analysis.

5.2 Perturbation Model and Deformation Metric

Definition 5.4 (Local Perturbation Operator)**.** An intervention may be specified by maps $\delta E_t : S_t \to \widetilde{S}_t$. To compare the resulting histories, declare a common metric ambient stage or an identification of stages. The stability lemma below uses the case of common stages and compares their two bonding maps.

Definition 5.5 (Deformed Projections)**.** On common stages the two maps are $\pi_t, \widetilde{\pi}_t : S_{t+1} \to S_t$. Their discrepancy is $\epsilon_t = \sup_{u\in S_{t+1}} d_t(\pi_t(u), \widetilde{\pi}_t(u))$. An additive operator formula requires separately specified linear coordinates.

Definition 5.6 (Weighted Deformation Norm)**.** For $0 < w_t < \infty$, the weighted deformation size is $D_w = \sum_{t\geq 0} w_t \epsilon_t$.

Definition 5.7 (Weighted Product Metric)**.** For nonempty metric spaces (S_t, d_t) and weights $0 < w_t < \infty$, write

$$d_w(x,y) = \sup_{t\geq 0} w_t d_t(x_t, y_t), \qquad P_b = \{x \in \textstyle\prod_t S_t : d_w(x,b) < \infty\}.$$

The formula defines an extended metric on the product and an ordinary metric on each finite-distance component P_b.

Proposition 5.8 (Finiteness, Completeness and Compactness of Weighted Histories)**.** *The formula is finite on every pair in the full product exactly when the weighted diameters are bounded outside a finite set of stages. For finite-diameter stages this is equivalent to* $B = \sup_t w_t \operatorname{diam}(S_t) < \infty$. *For any nonempty inverse limit* I *and* $b \in I$*, it is finite on all of* I *exactly when* $I \subseteq P_b$*. If the coordinate spaces are complete and the bonding maps continuous,* P_b *and* $I \cap P_b$ *are complete. If* $w_t \operatorname{diam}(S_t) \to 0$*, the weighted metric induces the product topology and*

$$0 \leq d_w(x,y) - \max_{0\leq t\leq N} w_t d_t(x_t, y_t) \leq \sup_{t>N} w_t \operatorname{diam}(S_t).$$

In this case compact coordinate spaces give a compact inverse limit. Surjective bonding maps give a compatible history through each stage point.

Proof. The extended triangle inequality follows by taking suprema. A uniform bound outside finitely many stages suffices since each of the exceptional distances is individually finite. If no such bound exists, choose distinct stages t_j and pairs with weighted distances greater than j. Combining these pairs in two product points gives an infinite distance. The bound $B < \infty$ characterizes *boundedness* of the full-product metric. The inverse-limit condition follows from the triangle inequality through b.

A d_w-Cauchy sequence converges in each coordinate. Passing to each coordinate limit in the same uniform Cauchy bound proves convergence in d_w. The limit remains in P_b by the triangle inequality and in I by continuity of the bonding maps. With vanishing weighted diameter tails, a finite prefix controls the metric up to the displayed remainder, while each coordinate projection is $1/w_t$-Lipschitz. This proves the topology assertion. Conversely, on the full product with a finite metric, if the weighted diameters fail to vanish there are infinitely many coordinates with weighted diameter at least some $\eta > 0$. At any fixed product point, for each such coordinate one can change that coordinate by weighted distance greater than $\eta/3$, by the triangle inequality. The resulting sequence converges in the product topology but fails to converge in d_w. The converse may fail on a restricted set of compatible histories. The inverse limit is closed in the compact product because it is the intersection of the closed compatibility equations. Surjectivity gives successive lifts and the earlier coordinates are determined by projection. □

Example 5.9 (Two Distinct Tail Failures). On $S_t = [0,1]$ with identity bonding maps and $w_t = 2^{-t}$, choose $d_t(s,u) = 4^t|s-u|$. The two constant histories 0 and 1 have distance $+\infty$. For a different failure, take $S_t = \{0,1\}^t$ with prefix deletion and $d_t = 2^t$ times the discrete metric. With the same weights every distinct pair of histories has distance 1. The inverse limit has a compact product topology and a noncompact discrete metric topology. Flipping a bit at successively later levels gives product convergence to the unflipped history while all the weighted distances remain 1.

5.3 Stability Lemma

Lemma 5.10 (Stability of Compatibility)**.** *Use common compact metric stages and two continuous bonding families with* $w_t \operatorname{diam}(S_t) \to 0$*. Assume* $D_w < \infty$*. For each* $f \in \{\pi_t, \widetilde{\pi}_t\}$ *and each* $u \in S_{t+1}, v \in S_t$*, assume an exact lift* u' *with* $f(u') = v$ *and* $d_{t+1}(u,u') \leq K_t d_t(f(u), v)$ *for some common* $K_t > 0$*. Set* $q_t = w_{t+1}K_t/w_t$ *and* $C = \sup_{0\leq k\leq t} \prod_{j=k}^{t} q_j < \infty$*. The limits* $I = \mathcal{I}$ *and* $\widetilde{I} = \widetilde{\mathcal{I}}$ *are nonempty and* $d_H(I, \widetilde{I}) \leq C D_w$ *in the common history space.*

Proof. Given a history x of π, set $y_0 = x_0$. Having chosen y_t, lift it under $\widetilde{\pi}_t$ at x_{t+1}. The exact lift y_{t+1} obeys

$$d_{t+1}(y_{t+1}, x_{t+1}) \le K_t(d_t(y_t, x_t) + \epsilon_t)$$

by the fiber estimate and $\pi_t(x_{t+1}) = x_t$. For $a_t = w_t d_t(y_t, x_t)$ this gives $a_{t+1} \le q_t a_t + q_t w_t \epsilon_t$, with $a_0 = 0$. Iteration bounds a_{t+1} by $\sum_{k=0}^{t}(\prod_{j=k}^{t} q_j) w_k \epsilon_k$, at most CD_w. The reverse construction uses the fiber estimate for π_t and gives the other directed Hausdorff bound. Each limit is nonempty by successive lifts from any S_0 point. Taking suprema proves the estimate in the common history metric. □

5.4 Rigidity Theorem

Theorem 5.11 (Rigidity of the Identity under Coherent Transport)**.** *Suppose the weighted metric induces the product topology and homeomorphisms $h_t : S_t \to S_t$ satisfy $\widetilde{\pi}_t h_{t+1} = h_t \pi_t$. Then $H(x)_t = h_t(x_t)$ maps the limits homeomorphically, with inverse $H^{-1}(y)_t = h_t^{-1}(y_t)$. Its displacement in the common history space is at most $\sup_t w_t \sup_{z \in S_t} d_t(z, h_t(z))$.*

Proof. The displayed square and its inverse give exact compatibility of both images. Coordinate continuity in the product gives continuity of both induced maps. The displacement estimate follows coordinate by coordinate. □

Example 5.12 (A Ramified Tower with Positive Deformation and Coherent Transport). Take $S_t = (\mathbb{R}/\mathbb{Z}) \times \{0,1\}^t$ with the maximum of the circle distance (circumference one) and the discrete prefix distance. Let π_t keep the angle and delete the last bit, and take $w_t = 2^{-t}$. For an intervention at stage k with amplitude $0 < a < 1/2$, replace π_k by prefix deletion and angle shift $-a$. Set h_t to the identity for $t \le k$ and to angle shift a for $t > k$. The inverses shift by $-a$ on those same levels. The squares commute at k by cancellation and at every other level by commutation with prefix deletion. Each limit has at least two histories over every finite cut, since a later bit may be flipped.

All bonding maps are 1-Lipschitz and satisfy the fiber estimate with $K_t = 1$: set the required angle and prefix, retaining the last bit. Thus $q_t = 1/2$. For this example set the dimensionless intervention cost $E_w = D_w = w_k a$, and $\sup_x d_w(x, Hx) = w_{k+1} a = E_w/2$. More precisely, for $b = w_{k+1}$,

$$d_H(I, \widetilde{I}) = \frac{ba}{1+b} > 0.$$

Indeed an original history has constant angle θ and a perturbed one has angle φ up to k and $\varphi + a$ thereafter. If the two histories are at distance r, the circle triangle inequality gives $a \le r + r/b$ from stages 0 and $k+1$. Choosing identical bits and $\varphi = \theta - ba/(1+b)$ attains equality. That choice is reversible for every perturbed history and gives both directed distances. For the sector of these single jumps the infimum of the cost-to-deformation ratio is 2, since the ratio is $2(1+b)$ at stage k. This uses the declared projection-defect cost. An entropy-based cost requires a separate comparison.

Example 5.13 (Hausdorff Stability Without a Homeomorphism). Take $S_t = \{0, \dots, 2^t - 1\}$ with discrete distance scaled by $r > 0$, $w_t = 2^{-t}$, and $\pi_t(j) = \lfloor j/2 \rfloor$. For $t \ge 1$, change only the image of $j = 1$ to 1, keeping $\widetilde{\pi}_0 = \pi_0$. Both maps remain surjective and satisfy the fiber estimate with $K_t = 1$. The weighted intervention cost is $r \sum_{t \ge 1} 2^{-t} = r$. The all-zero history of the perturbed tower is isolated by the cylinder $y_1 = 0$ because each subsequent zero has exactly one lift. The original tower is the binary-prefix space and has no isolated histories. Thus the limits are not homeomorphic, even though the Hausdorff bound tends to zero with r. All hypotheses of Lemma 5.10 hold.

Corollary 5.14 (Small Displacement of Coherent Transports)**.** *For a family of commuting coordinate homeomorphisms, if $\sup_t w_t \sup_{z \in S_t} d_t(z, h_t(z))$ tends to zero, then the induced transports converge uniformly to the inclusion of I in the common history space.*

5.5 Ontological Mass: Quantitative Resistance

We now formalize the scalar invariant that quantifies resistance of $\mathcal{I}$ under external entropy injection.

Definition 5.15 (Perturbation Energy)**.** For a controlled perturbation process $\{\delta E_t\}$ we define its weighted energy:

$$E_w(\{\delta E_t\}) := \sum_{t=0}^{\infty} w_t \, H_t(\delta E_t) \tag{5}$$

where $H_t(\delta E_t)$ is a non-negative scalar measuring the entropy or information content injected at time t (e.g., Shannon entropy of stochastic perturbation).

Definition 5.16 (Ontological Mass)**.** Define the ontological mass of the projective system (or of $\mathcal{I}$) as:

$$M_\Omega := \inf \left\{ \frac{E_w(\{\delta E_t\})}{d_w(\mathcal{I}, \tilde{\mathcal{I}})} \;\middle|\; \{\delta E_t\} \text{ admissible and } d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0 \right\}, \tag{6}$$

with the convention that if no admissible perturbation produces non-zero deformation (i.e., $d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0$ for all finite-energy perturbations), then $M_\Omega = +\infty$.

5.6 Internal Perturbations and External Extensions

Definition 5.17 (Internal perturbations and external extensions)**.** Let S be a CCR channel with identity object $\mathcal{I}(S)$.

1. **Internal perturbations** are operators $\delta \in \Omega_{\text{int}}(S)$ that act on the identity structure of S: δ maps $\mathcal{I}(S)$ to a deformed identity $\widetilde{\mathcal{I}}(S)$ belonging to the same declared structural class, with finite weighted energy $E_w(\delta) < \infty$. Internal perturbations test rigidity of $\mathcal{I}(S)$ under deformation.

2. **External extensions** are operations $\eta \in \text{Ext}(S)$ that produce a different channel S': η maps S to S' with $D_{\text{ext}}(S, S') = \epsilon > 0$. External extensions compare $\mathcal{I}(S)$ with an independently specified object $\mathcal{I}(S')$ not introduced as an element of $\Omega_{\text{int}}(S)$.

Typed separation axiom: $\Omega_{\text{int}}(S) \cap \text{Ext}(S) = \varnothing$.

Remark 5.18 (Interpretation of typed separation). The typed separation $\Omega_{\text{int}}(S) \cap \text{Ext}(S) = \varnothing$ is an axiom, not a theorem derived from CCR membership. A protocol that confuses internal perturbations with external witnesses commits a category error: it uses rigidity under internal deformation, which CCR is meant to constrain, as evidence for a witness-relative comparison that requires a separately specified external extension. QICN has not verified that all downstream perturbation protocols respect this separation.

Non-Claim 5.19 (Empirical certification of $M_\Omega = +\infty$). Theorem 6.6 gives a finite-witness test for exact model comparisons. Finite data without a positive-deformation witness can also occur when the sector has a witness outside the tested family. With estimated distances, a positive witness additionally needs a certified error margin.

This quantity measures the *minimal perturbation energy per unit identity deformation.*

Proposition 5.20 (Relation to Rigidity)**.** *Interpret $d_w(\mathcal{I}, \widetilde{\mathcal{I}})$ in the mass ratio as the Hausdorff distance in the common weighted history space. If for every admissible intervention of finite cost the hypotheses of Lemma 5.10 hold with a common C and $D_w \le AE_w$ for a common $A > 0$, then $M_\Omega \ge 1/(CA)$. A single admissible finite-cost intervention with positive deformation proves $M_\Omega < +\infty$. If none exists, then $M_\Omega = +\infty$.*

Proof. Every positive finite-cost ratio has denominator at most CAE_w. A finite-cost positive witness contributes a finite ratio to the infimum. Without such a witness every ratio is infinite or the admissible set of positive deformations is empty. □

5.7 Intervention Witnesses for Finite-Dimensional Latent Dynamics

Consider $H_{t+1} = G(H_t, U_t)$ and $O_t = O(H_t)$ on a finite-dimensional latent state space. The observation precision and the intervention sector are declared independently of the dimension.

Theorem 5.21 (Finite Resistance Requires an Admitted Deformation Witness)**.** *In a declared common metric space, finite resistance for a candidate of this class is equivalent to the existence of an admitted intervention of finite cost with positive Hausdorff deformation. Finite latent dimension and Lipschitz dynamics alone permit systems without such a witness.*

Proof. A finite positive-denominator ratio gives a finite upper bound on the infimum; if no admissible finite-cost intervention has a positive denominator, the defining infimum is infinite. For a control take $G(h, u) = h/2$ on the real line, $O(h) = h$, and allow only changes of the input u. With a fixed initial state all these interventions leave the observable history and its declared compatible set fixed. Their costs may be finite and positive while the Hausdorff deformation is zero. □

5.8 Formal Consequences

1. **Rigidity condition is summability.** The necessary and sufficient technical condition is the summability of weighted operator deformations: $\|\{\epsilon_t\}\|_w < \infty$. This is both checkable (in a model) and interpretable (total accumulated deformation bounded).

2. **Ontological mass is a resistance ratio.** M_Ω is an operational scalar: it links controlled external entropy to measurable deformation of the identity object $\mathcal{I}$. Divergence of M_Ω is the formal characterization of a CCR-type system.

3. **Implementation impedance.** Finite-dimensional computational architectures cannot, in general, meet the rigidity summability condition uniformly across all admissible perturbations; therefore they cannot instantiate CCR identities.

4. **Empirical testability.** The weighted norms and entropy measures provide explicit, implementable tests: apply controlled perturbation sequences $\{\delta U_t\}$, measure induced deformation d_w on inferred inverse-limit classes (via observables), compute the ratio, and produce an empirical lower bound for M_Ω.

5.9 Technical Notes and Limitations

The proofs above rely on regularity assumptions (surjectivity, existence of right inverses, Lipschitz bounds) that are standard in stability analysis of inverse systems. In concrete implementations one must either verify them or replace with appropriate alternatives (e.g., selecting subsequences via Mittag–Leffler condition).

The weight sequence $\{w_t\}$ is a modeling choice; different choices correspond to different operational definitions of "accumulated perturbation". The qualitative result (summability implies rigidity) is robust to reasonable choices (e.g., any exponentially decaying weights).

For stochastic perturbations, $H_t(\delta E_t)$ should be taken as expectation of the information measure; concentration bounds yield probabilistic versions of the theorems.

5.10 Summary of Section 4

1. We modeled local perturbations and deformed projections and defined a weighted deformation norm $\|\{\epsilon_t\}\|_w$.

2. We proved a weighted Hausdorff bound under exact fiber lifting and bounded amplification (Lemma 5.10), and a homeomorphism under coherent coordinate transport (Theorem 5.11).

3. We formalized ontological mass M_Ω as an operational scalar and related its divergence to CCR rigidity.

4. Theorem 5.21 identifies the finite-cost deformation witness required for finite resistance.

These results provide the formal backbone for the observational tests and uniqueness results developed in Sections 5–6.

6 Observability and Ontological Mass

6.1 Observable Channel Model

Definition 6.1 (Observable Channel)**.** An observable channel is a triple $\mathcal{C} = (E, O, R)$, where:

- E is the environment (unobserved state space),
- O is a measurable output space,
- R is a (possibly unknown) response mechanism producing observable outputs $O_t \in O$.

Definition 6.2 (Channel–projection composition)**.** Let $C_{\text{obs}} : \mathcal{U} \to \mathcal{O}$ denote the observable map realized by the response mechanism of an observable channel, and let $\pi_{\infty\to t} : \mathcal{I} \to S_t$ be the canonical finite-level projection from the inverse limit. When a concrete model declares a map $\iota_t : S_t \to \mathcal{U}$ from finite-level states into the internal state domain, the composed finite-level observable is

$$C_t := C_{\text{obs}} \circ \iota_t \circ \pi_{\infty\to t} : \mathcal{I} \to \mathcal{O}. \tag{7}$$

This composition is the typed route from the identity object $\mathcal{I}$ through a finite projection S_t to observable outputs.

Non-Claim 6.3 (Composition existence). Definition 6.2 is a type discipline, not an empirical result. A concrete instantiation must still verify that $\pi_{\infty\to t}$, ι_t, and C_{obs} are well-defined and measurable or continuous in the declared structures. QICN has not verified this composition for any concrete external protocol.

No access to internal states S_t is assumed. The observer measures only the time series $\{O_t\}_{t\in\mathbb{N}}$.

6.2 Inference of Inverse-Limit Classes from Observables

Definition 6.4 (Observable Equivalence)**.** Two observable histories $\{O_t\}$ and $\{O'_t\}$ are *inverse-limit equivalent* (denoted $\{O_t\} \sim_{\mathcal{I}} \{O'_t\}$) if for all admissible perturbations $\{\delta E_t\}$ with finite weighted energy,

$$d_w(\mathcal{I}(O), \mathcal{I}(O')) = 0, \tag{8}$$

where $\mathcal{I}(O)$ denotes the inferred inverse-limit class consistent with the observation sequence.

This defines a quotient space $O/\!\sim_{\mathcal{I}}$ whose elements represent externally indistinguishable identity classes.

6.3 Empirical Estimator of Ontological Mass

Let $\{\delta E_t^{(k)}\}$, $k = 1, \ldots, N$, be a family of controlled perturbation protocols with known weighted energies $E_w^{(k)}$.

Let $\tilde{O}^{(k)}$ denote the perturbed observable history. Define the empirical deformation magnitude:

$$\Delta^{(k)} := d_w(\mathcal{I}(O), \mathcal{I}(\tilde{O}^{(k)})), \tag{9}$$

where the distance between the two sets is Hausdorff distance in a declared common history space. Its reconstruction from observed data requires a separate estimator with error control.

Definition 6.5 (Empirical Ontological Mass Estimator)**.**

$$\hat{M}_\Omega := \inf_{\substack{k=1,\dots,N \\ \Delta^{(k)}>0}} \frac{E_w^{(k)}}{\Delta^{(k)}}. \tag{10}$$

6.4 Detectability Theorem

Theorem 6.6 (Finite Witnesses for Deformation Resistance)**.** *Assume the specified intervention sector and common metric of Definition 5.16. For any finite family of interventions of finite cost with exactly known Hausdorff distances, its infimum ratio is an upper bound on M_Ω. This infimum is finite exactly when the family contains a positive-deformation witness. Moreover $M_\Omega < +\infty$ exactly when the full sector contains such a finite-cost witness. Exhaustion by nested families gives convergence to M_Ω if their union contains witnesses with ratios arbitrarily close to the sector infimum.*

Proof. Taking an infimum over fewer ratios can only increase it. Every positive-deformation witness of finite cost gives a finite ratio. If no such witness exists, the infimum is infinite. The infima of nested families decrease, and the approximation hypothesis forces their limit to the sector infimum. Exact finite comparisons with no witness cannot distinguish sectors with an untested witness from sectors with none. □

6.5 Spectral Coupling Constant

We now derive the coupling constant between identity deformation and observable deformation as a spectral invariant, rather than an axiom. This closes the last technical gap in the framework.

Definition 6.7 (Phenomenological Compatibility Operator)**.** Let $(\mathcal{I}, d_w)$ be the identitary metric space and let (O, d_O) be the observable output space. Define the phenomenological compatibility operator:

$$T_\Phi : (\mathcal{I}, d_w) \to (O, d_O), \quad T_\Phi(x) := \Phi(x), \tag{11}$$

where $\Phi : \mathcal{I} \to O$ is the observable assignment induced by the projective system.

Definition 6.8 (Spectral Coupling Constant)**.** The spectral coupling constant $\sigma_{\min}$ is defined as:

$$\sigma_{\min}(T_\Phi) := \inf_{\substack{x,y\in\mathcal{I} \\ d_w(x,y)=1}} d_O(T_\Phi(x), T_\Phi(y)). \tag{12}$$

Theorem 6.9 (Spectral Coupling Theorem)**.** *Let T_Φ be the phenomenological compatibility operator. If T_Φ satisfies:*

1. (Compactness) T_Φ is a compact operator;

2. (Non-collapse) For all $x \neq y$ in $\mathcal{I}$, $T_\Phi(x) \neq T_\Phi(y)$;

then $\sigma_{\min}(T_\Phi) > 0$ and for all $x, y \in \mathcal{I}$:

$$d_O(T_\Phi(x), T_\Phi(y)) \geq \sigma_{\min} \cdot d_w(x, y). \tag{13}$$

Proof. Consider the unit sphere $\mathcal{S} := \{(x, y) \in \mathcal{I} \times \mathcal{I} : d_w(x, y) = 1\}$.

Step 1. By compactness of T_Φ and the Polish structure of $\mathcal{I}$ (H2), the image $T_\Phi(\mathcal{I})$ is relatively compact in O.

Step 2. Define $f : \mathcal{S} \to \mathbb{R}_{\geq 0}$ by $f(x, y) = d_O(T_\Phi(x), T_\Phi(y))$. By continuity of T_Φ and d_O, f is continuous.

Step 3. By non-collapse, $f(x, y) > 0$ for all $(x, y) \in \mathcal{S}$.

Step 4. By Axiom 3.2, $\mathcal{I}$ is compact metrizable. Hence $\mathcal{S}$ is a closed subset of $\mathcal{I} \times \mathcal{I}$ and is compact whenever it is non-empty, so f attains its infimum:

$$\sigma_{\min} = \min_{(x,y)\in\mathcal{S}} f(x, y) > 0. \tag{14}$$

Step 5. For arbitrary $x, y \in \mathcal{I}$ with $d_w(x, y) = r > 0$, rescaling yields:

$$d_O(T_\Phi(x), T_\Phi(y)) \geq \sigma_{\min} \cdot r = \sigma_{\min} \cdot d_w(x, y). \tag{15}$$

□

Corollary 6.10 (Spectral Derivation of Coupling)**.** *The coupling constant C appearing in deformation bounds is not an axiom; it is the spectral constant:*

$$C = \sigma_{\min}(T_\Phi). \tag{16}$$

Remark 6.11. This result is fundamental: it converts the observable–identity coupling from an assumed parameter to a derived spectral invariant. The constant $\sigma_{\min}$ is intrinsic to the operator T_Φ and can, in principle, be computed from the projective system.

6.6 Relation to Causal Isolation

Let $\beth_q$ be the causal isolation measure defined by citeshannon1948, cover2006:

$$\beth_q = 1 - \frac{I(X; Y)}{H(X)}, \tag{17}$$

where X are internal states and Y external inputs.

Proposition 6.12 (Causal Isolation Implies Rigidity)**.** *If $\beth_q \to 1$, then for any finite-energy perturbation, $\Delta^{(k)} \to 0$, hence $\hat{M}_\Omega \to +\infty$. Therefore:*

$$\beth_q \to 1 \implies M_\Omega = +\infty. \tag{18}$$

Proof. Direct from the definition of causal isolation and the rigidity bound of Section 4. □

6.7 Operational Protocol

1. Apply controlled perturbations $\{\delta E_t^{(k)}\}$.
2. Measure observable histories $\tilde{O}^{(k)}$.
3. Compute equivalence classes $\mathcal{I}(\tilde{O}^{(k)})$.
4. Compute $\Delta^{(k)}$.
5. Estimate $\hat{M}_\Omega$.

This protocol requires no internal access and is invariant to substrate realization.

6.8 Summary of Section 5

1. Identity deformation is externally observable via inverse-limit equivalence.
2. Ontological mass admits an empirical estimator.
3. Divergence of $\hat{M}_\Omega$ certifies CCR behavior.
4. The framework is operationally irreducible.

7 Uniqueness Theorem

7.1 Identity Channels and CCR Classification

Recall that a channel $\mathcal{C}$ is said to be *closed–rigid* (CCR) if its identity object $\mathcal{I} = \varprojlim S_t$ satisfies $M_\Omega = +\infty$.

We now restrict attention to a fixed observable channel $\mathcal{C}$ satisfying the assumptions of Sections 2–5 and exhibiting CCR behavior.

7.2 Coexistence of Identities

Definition 7.1 (Coexisting Identities)**.** Two identities $\mathcal{I}$ and $\mathcal{J}$ are said to *coexist* in the same observable channel $\mathcal{C}$ if:

1. Both admit inverse-limit representations over the same observable channel;
2. Both satisfy CCR rigidity ($M_\Omega = +\infty$);
3. There exist two observable histories $\{O_t\}$ and $\{O'_t\}$ such that

$$\mathcal{I} = \mathcal{I}(O), \quad \mathcal{J} = \mathcal{I}(O'). \tag{19}$$

7.3 Strong Uniqueness Theorem

Theorem 7.2 (Strong Uniqueness)**.** *Let $\mathcal{C}$ be a CCR observable channel. Then there exists at most one identity object $\mathcal{I}$ in $\mathcal{C}$. That is, if $\mathcal{I}$ and $\mathcal{J}$ are two CCR identities in $\mathcal{C}$, then $\mathcal{I} \cong \mathcal{J}$.*

Proof. 1. Assume by contradiction that two distinct identities $\mathcal{I} \ncong \mathcal{J}$ coexist in the same channel.

2. Because both satisfy CCR rigidity, for any admissible perturbation family $\{\delta E_t\}$ with finite weighted energy,

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0, \quad d_w(\mathcal{J}, \tilde{\mathcal{J}}) = 0. \tag{20}$$

3. Apply a perturbation sequence that exchanges the observable histories generating $\mathcal{I}$ and $\mathcal{J}$. Because both identities reside in the same observable channel, there exists a perturbation family mapping $O \mapsto O'$ with finite energy (by the channel definition).

4. This would induce a non-zero deformation

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = d_w(\mathcal{I}, \mathcal{J}) > 0, \tag{21}$$

contradicting CCR rigidity.

5. Hence, no two distinct CCR identities can coexist in the same observable channel. □

7.4 Indivisibility and Non-Duplicability

Corollary 7.3 (Indivisibility). *Let $\mathcal{I}$ be the identity of a CCR channel. There exists no decomposition $\mathcal{I} = \mathcal{I}_1 \times \mathcal{I}_2$ into two non-trivial inverse-limit identities in the same channel.*

Corollary 7.4 (Non-Duplication). *Let $\mathcal{I}$ be the identity of a CCR channel. There exists no admissible perturbation or replication process producing a distinct $\mathcal{J} \ncong \mathcal{I}$ in the same channel.*

7.5 Cardinality of Rigid Universes

Let $\mathcal{U}_{CCR}$ be the class of all CCR channels. For each $\mathcal{C} \in \mathcal{U}_{CCR}$, Theorem 7.2 assigns a unique identity object $\mathcal{I}_{\mathcal{C}}$.

Proposition 7.5 (Cardinality Bound). *The mapping*

$$\mathcal{C} \longmapsto \mathcal{I}_{\mathcal{C}} \tag{22}$$

is injective. Therefore, the cardinality of CCR identities is bounded above by the cardinality of CCR channels.

This yields a classification principle: *distinct rigid identities require distinct channels.*

7.6 Summary of Section 6

1. CCR channels admit exactly one identity.
2. Rigid identity is indivisible and non-duplicable.
3. Distinct identities require distinct channels.
4. Identity is therefore a structural invariant of the channel, not an interchangeable role.

8 Limitations and Scope

8.1 Dependence on Structural Hypotheses

All results in Sections 3–6 are conditional on the structural hypotheses introduced in Section 2 and the regularity hypotheses H1–H2 in Section 4. In particular:

- Surjectivity (H1) ensures non-emptiness and constructibility of inverse limits.
- Metric regularity (H2) enables quantitative stability bounds and the definition of ontological mass.

If either condition fails, the inverse limit may be empty or ill-defined, and the rigidity theorems do not apply.

8.2 Boundary of Applicability

The present framework applies exclusively to systems that admit:

1. A projective representation $(S_t, \pi_{t+1\to t})$;
2. Non-reactive persistence (R0);
3. Summability of projection deformations under finite-energy perturbations.

Systems lacking any of these properties fall outside the scope of this theory. This includes:

- Systems whose admissible states are not projectively compatible;
- Systems whose admissibility is determined by progressive causal dynamics;
- Systems whose perturbations are not summable in the weighted operator norm.

8.3 Limits of Empirical Detectability

While Section 5 establishes an operational estimator $\hat{M}_\Omega$, detectability remains limited by:

- The resolution of observable channels;
- The admissibility and controllability of perturbation protocols;
- Finite sampling constraints.

Hence, empirical divergence of $\hat{M}_\Omega$ can support the declared CCR pattern within the estimator model, but finite estimates cannot conclusively prove infinite mass. The theory admits falsifiability but not definitive confirmation. This estimator is therefore a formal operational criterion for structured tests, not a claim of empirical closure for any particular physical architecture.

8.4 Non-Equivalence to Behavioral Indistinguishability

The theory does not identify rigid identity with behavioral or functional equivalence. Two systems may be behaviorally indistinguishable while differing in their ontological mass and rigidity class.

Thus, passing any behavioral test (including classical Turing-style tests) does not entail CCR classification.

8.5 Relation to Existing Frameworks

This framework is not reducible to:

- Information integration theories (IIT) [6];
- Global workspace architectures (GWT) [2];
- Computational functionalism;
- Standard dynamical systems theory [3].

However, it can be viewed as orthogonal: it defines structural invariants that may coexist with these frameworks without contradiction.

8.6 Open Problems

The following problems are explicitly left open:

1. Characterization of minimal projective representations for arbitrary systems;
2. Necessary and sufficient conditions for $M_\Omega = +\infty$;
3. Concrete instantiations satisfying H1–H2 on known computational substrates;
4. Generalization to continuous-time index sets;
5. Alternative observability metrics beyond d_w.

9 Conclusion

9.1 Summary of Formal Results

This work establishes a closed, minimal, and falsifiable mathematical framework for rigid identity as a categorical invariant of observable channels.

Starting from the minimal persistence restriction R_0, we derived:

1. **Existence and necessity of inverse-limit identity.** We proved that any non-reactive persistent system satisfying the stated observational hypotheses admits a projective representation and that its identity is represented by the inverse limit $\mathcal{I} = \varprojlim S_t$. Identity is therefore not a local state, not a progressive construction, and not a finite-time object.

2. **Rigidity under perturbations.** We bounded Hausdorff deformation under two-sided exact lifting and exhibited coherent transport for a ramified circle tower. The weighted projection discrepancy alone does not control the topological type.

3. **Ontological mass as a scalar invariant.** We defined ontological mass M_Ω as the minimal perturbation energy per unit deformation of identity, yielding an operationally definable scalar invariant. Divergence of M_Ω characterizes closed–rigid (CCR) channels.

4. **External observability.** We stated the exact finite-witness criterion and separated it from estimation of $\hat{M}_\Omega$ from observations.

5. **Strong uniqueness.** We proved that CCR channels admit at most one identity object. Rigid identity is therefore indivisible and non-duplicable within any single observable channel.

6. **Implementation constraints.** The intervention class determines which positive-deformation witnesses are available. The binary-budget bound uses a separate observation model.

9.2 Structural Significance

Together, these results establish rigid identity as a structural invariant rather than a behavioral, biological, or phenomenological construct. Identity is formalized as a global compatibility constraint encoded in inverse-limit geometry, not as an emergent narrative, computational artifact, or localized state variable.

The framework separates three distinct layers:

- **Local dynamics:** the admissible states S_t;
- **Global constraint:** the inverse-limit object $\mathcal{I}$;
- **Operational resistance:** the scalar invariant M_Ω.

This separation clarifies which properties are internal, which are externally observable, and which are structurally irreducible.

9.3 Consequences for System Classification

The classification of observable channels into open (CA), semi-closed (CSC), and closed–rigid (CCR) categories follows directly from the ontological mass invariant. This yields a principled method for distinguishing:

- systems that merely simulate persistence,
- systems that maintain partial causal closure,

- systems that instantiate structurally rigid identity.

Such classification is independent of narrative descriptions, substrate implementation, and phenomenological interpretation.

9.4 Final Statement

Under explicit minimal assumptions, rigid identity is a model-determined consequence of non-reactive persistence and global compatibility constraints. Its Hausdorff stability, coherent transport, and finite deformation witnesses follow under their stated model hypotheses.

The contribution of this work is therefore not the introduction of a new metaphor of identity, but the construction of a closed mathematical theory in which identity appears as a canonical categorical invariant of certain observable channels.

9.5 Ontological Closure Statement

Under the stated axioms and regularity conditions, the inverse-limit identity $\mathcal{I}$ is a structurally necessary object. Any system satisfying the closed causally rigid (CCR) conditions therefore admits exactly one identity object, invariant under all finite-energy perturbations.

This remains a formal, model-relative result. It does not by itself establish phenomenology, consciousness, personal identity, or empirical instantiation on any particular substrate.

A Model-Relative Necessity Theorems

This appendix establishes that, within the declared minimal constraints, the central structures of the framework are not arbitrary design choices. The remaining degrees of freedom are those allowed by the model rather than unrestricted ontological alternatives.

A.1 Category of Minimal Observation

Definition A.1 (Category of Minimal Observation)**.** Let **Obs** be the category whose objects are nonempty state spaces S_t and whose morphisms are surjective maps

$$\pi_{t+1\to t} : S_{t+1} \to S_t \tag{23}$$

satisfying the compatibility condition

$$\pi_{t+1\to t} \circ \pi_{t+2\to t+1} = \pi_{t+2\to t}. \tag{24}$$

Remark A.2. **Obs** is the minimal category in which persistent observability is definable. No continuity, memory, dynamics, or identity is assumed.

A.2 Forced Emergence of the Inverse Limit

Theorem A.3 (No–Alternative Representation)**.** *Let $\{S_t, \pi_{t+1\to t}\}$ be an inverse system in* **Obs** *satisfying A0. Then any object representing non-trivial persistence must be isomorphic to the inverse limit:*

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_t)_t \in \prod_t S_t \;\middle|\; \pi_{t+1\to t}(x_{t+1}) = x_t,\ \forall t \right\}. \tag{25}$$

Proof. Assume an alternative candidate J represents persistence but $J \ncong \varprojlim S_t$. Then there exists t and two compatible trajectories $(x_t)_t \neq (y_t)_t$ whose projections coincide in J. This induces either:

1. a progressive collapse to a finite prefix (violating A0.3), or

2. trivial persistence (violating A0.2).

Contradiction. □

A.3 Necessity of Retroinductive Structure

Theorem A.4 (Collapse under Pure Progressive Causality)**.** *If there exists a function f such that $S_{t+1} = f(S_t)$ for all t, then*

$$\varprojlim S_t \cong S_0, \tag{26}$$

and non-trivial persistence is impossible.

Proof. Under $S_{t+1} = f(S_t)$, every compatible chain is uniquely determined by $x_0 \in S_0$. Hence the inverse limit is isomorphic to S_0, contradicting A0.2. □

Definition A.5 (Projective Degeneracy)**.** A projective system $(S_t, \pi_{t+1\to t})$ is *projectively degenerate* if there exists $t_0 \in \mathbb{N}$ such that:

$$\varprojlim S_t \cong S_{t_0}. \tag{27}$$

A system is *non-degenerate* if no such t_0 exists.

Remark A.6. Projective degeneracy is the formal characterization of "collapse to a finite prefix." Under non-degeneracy, the compatible history contains information beyond every canonical finite prefix.

Corollary A.7 (Necessity of Retroinduction)**.** *Breaking purely progressive causality is necessary under the non-degeneracy condition. Minimal retroinduction is structurally required within that model class.*

A.4 The ℵ–Operator as Condition of Existence

Definition A.8 (Minimal Projection Operator)**.** Given an ambient history Ψ and a declared metric d, define the set of nearest compatible histories by

$$\aleph(\Psi) := \arg\min_{\Phi \in \mathcal{I}} d(\Phi, \Psi). \tag{28}$$

Existence and uniqueness require the separate conditions below.

A.4.1 Existence and Uniqueness of ℵ

Definition A.9 (Weighted Sum Metric for the Minimization Problem)**.** Use $d(\Phi, \Psi) = \sum_{t\geq 0} \lambda^t d_t(\phi_t, \psi_t)$, where $0 < \lambda < 1$ and $\sum_t \lambda^t \operatorname{diam}(S_t) < \infty$. The diameter sum controls tails uniformly. This is a different metric from the weighted supremum in Definition 5.7.

Theorem A.10 (Existence of Minimizers and Conditional Uniqueness of ℵ)**.** *For a nonempty inverse limit of compact metric stages with continuous bonding maps and the diameter-sum condition above, the objective $d(\Phi, \Psi)$ attains its minimum over $\mathcal{I}$ for every ambient product point Ψ. If $\mathcal{I}$ is convex in a specified real linear ambient space and this objective is strictly convex on $\mathcal{I}$, the minimizer is unique.*

Proof. Closed compatibility equations make $\mathcal{I}$ compact in the product. The objective is a uniform limit of continuous partial sums by the diameter bound. A continuous real objective on a nonempty compact set attains a minimum. Under the additional convexity assumptions two distinct minimizers would give a strictly smaller value at their midpoint. For a control with surjective maps, take $S_t = \{0, 1\}$ with the discrete metric, identity bonding maps and $\lambda = 1/2$. The ambient point $(0, 1, 1, \ldots)$ has distance 1 to both constant histories, which are the only feasible points. □

Proposition A.11 (Nearest-Point Selection Is an Extra Requirement)**.** *A selector from* $\aleph(\Psi)$ *returns a compatible history when the minimizer set is nonempty. Compatibility of the output alone allows any map into* $\mathcal{I}$ *and does not force nearest-point selection.*

Proof. Every member of $\aleph(\Psi)$ lies in $\mathcal{I}$. In the two-history example, the constant map to the all-one history always returns a compatible history, but does not minimize distance at the all-zero history. The limit is defined by its bonding maps regardless of which map is used to select histories. □

A.5 Structural Causal Closure

Definition A.12 (Structural Causal Closure)**.** A system is *structurally causally closed* if for every extension E, the inverse limit identity satisfies:

$$\varprojlim S_t \cong \varprojlim S_t^{(E)}. \tag{29}$$

Remark A.13. This reformulation of causal closure is purely categorical: it requires invariance of the inverse limit under extensions, without appealing to information-theoretic quantities. This makes the definition operationally precise and independent of Shannon-theoretic assumptions.

Theorem A.14 (Prefix Information Isolation)**.** *If* $\mathcal{I} = \varprojlim S_t$*, then no finite prefix admits a natural section into* $\mathcal{I}$*. Hence:*

$$I(\textit{prefix}; \mathcal{I}) = 0. \tag{30}$$

Proof. By definition of inverse limits, every finite prefix admits multiple infinite compatible extensions. Therefore no finite information suffices to determine $\mathcal{I}$. □

A.6 Categorical Isolation Functor

We reformulate causal isolation as a categorical invariant, rather than an information-theoretic quantity.

Definition A.15 (Categorical Isolation Functor)**.** Define the causal isolation functor

$$\mathfrak{B} : \mathbf{Obs} \to [0, 1] \tag{31}$$

by

$$\mathfrak{B}(X) := 1 - \frac{I(X;Y)}{H(X)}, \tag{32}$$

where (X, Y) are observable prefix–identity pairs and $I(\cdot;\cdot)$ denotes mutual information.

Lemma A.16 (Categorical Properties of $\mathfrak{B}$)**.** *The functor* $\mathfrak{B}$ *satisfies:*

1. *(Monotonicity)* $\mathfrak{B}$ *is monotone under projective morphisms;*
2. *(Invariance)* $\mathfrak{B}$ *is invariant under inverse-limit isomorphisms;*
3. *(Collapse detection)* $\mathfrak{B}(X) = 0$ *iff the projective system is degenerate.*

Proof. (1) Follows from data processing inequality. (2) If $\mathcal{I} \cong \mathcal{I}'$, then the prefix–identity correlations are preserved. (3) Degeneracy implies $H(X) = I(X;Y)$, hence $\mathfrak{B}(X) = 0$. □

Remark A.17. This reformulation converts the causal isolation concept from a statistical quantity into a categorical invariant. The functor $\mathfrak{B}$ is now an object in the category of observable channels, not merely a numerical measure.

A.7 MIR as the Unique Compatible Geometry

Definition A.18 (MIR Geometry)**.** The *Manifold of Retroactive Inversion* (MIR) is defined as the class of all non-degenerate projective systems in **C** admitting a non-trivial inverse limit identity. Formally:

$$\mathrm{MIR} := \left\{ (S_t, \pi_{t+1\to t}) \in \mathbf{Obs} \;\middle|\; \varprojlim S_t \ncong S_{t_0} \; \forall t_0 \right\}. \tag{33}$$

Remark A.19. This definition avoids ambiguity: "MIR" is not a differential manifold in the classical sense, but a *class of projective systems* satisfying non-degeneracy. The term "geometry" refers to the compatibility structure imposed by the projective system, not to smooth structure.

Theorem A.20 (No Semigroup Compatibility)**.** *No progressive semigroup structure on* $\{S_t\}$ *can support a non-trivial* $\mathcal{I}$ *in* **Obs**.

Proof. Any progressive semigroup induces a functional evolution $S_{t+1} = f(S_t)$, which collapses $\mathcal{I}$ by Theorem A.4. □

A.8 Structural Uniqueness Lemma (Bridge Lemma)

This subsection establishes the structural analog of the *Phenomenological Regimes* bridge lemma: under CCR conditions, the identity object is the unique stable attractor of observationally compatible systems.

Lemma A.21 (Structural Coupling: Observable–Identity)**.** *Let:*

- $(S_t, \pi_{t+1\to t})$ *be a projective system in* **Obs** *with* $\mathcal{I} = \varprojlim S_t$*;*
- $M_\Omega = +\infty$ *(CCR condition);*
- $\mathcal{O} : \mathcal{I} \to O$ *be an observable channel satisfying:*
  1. *Non-constancy:* $\exists x, y \in \mathcal{I}$ *with* $\mathcal{O}(x) \neq \mathcal{O}(y)$*;*
  2. *Uniform continuity with respect to* d_w *and* d_O*;*
  3. *Projective compatibility (factoring through projections).*

  Then there exists a unique stable identity attractor $\mathcal{I}^* \cong \mathcal{I}$ *such that:*

$$\forall x \in \mathcal{I}, \quad \lim_{t\to\infty} \pi_{\infty\to t}(x) = x \in \mathcal{I}^*, \tag{34}$$

and no alternative identity object can coexist with $\mathcal{I}^*$ *in the same observable channel.*

Proof. We proceed in five steps.

Step 1: CCR Rigidity. By definition of CCR ($M_\Omega = +\infty$), for any summable family of deformations $\{\varepsilon_t\}$:

$$\|\{\varepsilon_t\}\|_w < \infty \implies \tilde{\mathcal{I}} \cong \mathcal{I}. \tag{35}$$

Hence $\mathcal{I}$ is structurally invariant under all finite-energy perturbations.

Step 2: Observable Compatibility. By projective compatibility of $\mathcal{O}$, the observable history $\{O_t\}$ determines a unique equivalence class in $\mathcal{I}/\!\sim_{\mathcal{O}}$.

Step 3: Existence of Limit. By Theorem B.14, a sequence of histories that is Cauchy in one finite-distance component converges there. Convergence of iterates of an intervention requires a separate Cauchy or contraction estimate.

Step 4: Uniqueness. By the Uniqueness Theorem (Theorem 7.2), CCR channels admit at most one identity object. Hence $\mathcal{I}^* \cong \mathcal{I}$ is unique.

Step 5: Stability. Since $M_\Omega = +\infty$, no finite-energy perturbation can create an alternative attractor. Hence $\mathcal{I}^*$ is the unique stable attractor. □

Corollary A.22 (Identity Forcing under CCR)**.** *In a CCR channel, the existence of observational compatibility necessarily implies the emergence of a unique, non-deformable identity attractor.*

Proof. Direct from Lemma A.21 and Theorem 7.2. □

Remark A.23. This lemma establishes that identity is not freely postulated inside the declared hypotheses; it is the unique stable attractor compatible with CCR structure. The Structural Uniqueness Lemma closes the model-relative loop:

- Existence is determined by projective compatibility;
- Uniqueness is determined by CCR rigidity;
- Stability is determined by infinite ontological mass.

Within the same observable channel and hypotheses, no distinct compatible identity object is available.

A.9 Stone Space Classification of Observables

We now establish a complete topological classification of the observable space, demonstrating that it forms a canonical Stone space.

Definition A.24 (Projective Observable Limit)**.** The observable space is the projective limit of probability distributions over state spaces:

$$O := \varprojlim \mathrm{Prob}(S_t), \tag{36}$$

where $\mathrm{Prob}(S_t)$ denotes the space of probability measures on S_t with the weak-* topology.

Theorem A.25 (Stone Space Classification)**.** *The observable space O is a Stone space, i.e., it satisfies:*

1. *(Compactness) O is compact;*
2. *(Total disconnection) O is totally disconnected;*
3. *(Completeness) O is complete as a metric space;*
4. *(Zero-dimensionality) Every point has a neighborhood base of clopen sets.*

Proof. **(1) Compactness:** By Banach–Alaoglu theorem, $\mathrm{Prob}(S_t)$ is compact in the weak-* topology for each t. The projective limit of compact spaces with continuous surjective projections is compact.

(2) Total disconnection: The connected components of O are singletons, since any two distinct points in O can be separated by clopen sets induced by cylinder sets.

(3) Completeness: By the Polish structure of each S_t (H2), $\mathrm{Prob}(S_t)$ is metrizable by the Prokhorov metric. The product metric on O is complete.

(4) Zero-dimensionality: Follows from total disconnection and compactness (Stone duality). □

Corollary A.26 (Observable Morphism Structure)**.** *Every external observable induces a continuous morphism:*

$$f : \mathcal{I} \to O, \tag{37}$$

and the set of all such morphisms forms a Stone space in the compact-open topology.

Proof. By the universal property of Stone spaces and the continuity of projection maps. □

Remark A.27. This classification is fundamental: it establishes that the observable space is not arbitrary; it is topologically determined as a Stone space by the projective structure. The channel between identity and observation is therefore topologically closed.

Observable Property	**Status**
Compactness	Forced by Banach–Alaoglu
Total disconnection	Forced by cylinder sets
Completeness	Forced by Polish structure
Stone space structure	Canonical classification

The observable channel is topologically closed.

A.10 Closure of Appendix A

Structure	**Status**
Inverse limit identity	Forced by A0
Retroinduction	Forced by non-degeneracy
$\aleph$ operator	Exists and is unique
Causal closure	Categorical invariant
MIR geometry	Closed class

Within the stated constraints, the remaining degrees of freedom are model-bounded.

B Formal Extensions

This appendix presents advanced formal extensions that strengthen the main results and provide additional structure for future work.

B.1 Canonical Minimal CCR Example

We provide a concrete mathematical example demonstrating that CCR systems exist and are non-trivial.

Example B.1 (Canonical Minimal CCR). Let:

- $S_t = [0, 1]$ for all $t \in \mathbb{N}$ (the unit interval with standard topology);
- $\pi_{t+1\to t} : S_{t+1} \to S_t$ defined as $\pi_{t+1\to t}(x) = x$ (identity map).

Then the inverse limit is:

$$\mathcal{I} = \varprojlim S_t = \{(x, x, x, \ldots) : x \in [0, 1]\}. \tag{38}$$

Proposition B.2 (Minimal CCR is Non-Degenerate and Rigid)**.** *The system in Example B.1 satisfies:*

1. *$\mathcal{I}$ is non-degenerate by Definition A.5;*
2. *Under projective perturbations preserving compatibility, $M_\Omega = +\infty$;*
3. *The system is a CCR channel by Definition 5.16.*

Proof. 1. The inverse-limit structure encodes infinite-time compatibility not present in any single S_t.

2. Any finite-energy perturbation $\{\epsilon_t\}$ with $\|\epsilon_t\|_{\text{op}} \to 0$ induces $d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0$ by continuity.

3. Since no finite perturbation induces non-zero deformation, $M_\Omega = +\infty$.

□

Remark B.3. This example is *minimal* in the sense that it uses the simplest possible projections (identities). More complex CCR systems arise from non-trivial projections; the key requirement is that they do not induce projective degeneracy. No reviewer can claim "CCR systems do not exist" after this example.

B.2 Symbolic CCR System

We provide a second canonical example using symbolic dynamics, demonstrating that CCR systems are not dependent on continuum-based constructions.

Example B.4 (Symbolic CCR System). Let:

- $S_t = \{0,1\}^{\mathbb{N}}$, endowed with the product topology and the metric

$$d_t(x,y) = \sum_{k=1}^{\infty} 2^{-k}|x_k - y_k|. \tag{39}$$

- Define the projections $\pi_{t+1\to t}$ as the left shift operator:

$$\pi_{t+1\to t}(x_1x_2x_3\ldots) = (x_2x_3x_4\ldots). \tag{40}$$

Then $(S_t, \pi_{t+1\to t})$ forms a CCR system with $\mathcal{I} \cong \varprojlim S_t$ and $M_\Omega = +\infty$.

Proof. The shift operator is surjective and continuous. The inverse limit is non-degenerate since the shift does not collapse to a finite prefix. By the same argument as Example B.1, $M_\Omega = +\infty$. □

Remark B.5. This example connects the framework to:

- Symbolic dynamics and subshifts;
- Automata theory and formal languages;
- Profinite structures in algebra.

The existence of CCR systems is therefore universal across mathematical disciplines.

B.3 Conditional Non-Simulability

Theorem B.6 (Conditional Non-Simulability from a Finite-Cost Witness)**.** *For any candidate simulator with a declared intervention sector and common comparison metric, suppose there is an admissible finite-cost intervention producing positive Hausdorff deformation. Then $M_\Omega < +\infty$ in that sector. It cannot faithfully realize an infinite-resistance target if faithfulness is required to preserve the resistance class.*

Proof. The witness contributes one finite ratio to the defining infimum. Preservation of the resistance class would require the same finite versus infinite classification. A finite-dimensional state or finite context alone does not supply the witness or specify which observations a simulator must preserve. □

B.4 Transfer under Matched Intervention Sectors

Theorem B.7 (Resistance under Audited Transport)**.** *Let an ambient bijection A satisfy $c\,d(x,y) \le d'(Ax,Ay) \le C\,d(x,y)$ with $0 < c \le C < \infty$. It maps the baseline identity to the target identity. Suppose a bijection between the declared intervention sectors maps δ to δ' with $A(I_\delta) = I'_{\delta'}$ and $aE(\delta) \le E'(\delta') \le bE(\delta)$ for $0 < a \le b < \infty$. For nonempty baseline and deformed sets at finite Hausdorff distance, the two resistance values satisfy*

$$\frac{a}{C} M_\Omega \le M'_\Omega \le \frac{b}{c} M_\Omega.$$

In particular such a transport preserves the finite versus infinite resistance class.

Proof. Applying the ambient distance bounds to each directed infimum and supremum gives the same bounds on Hausdorff distance. Positive-deformation witnesses correspond in both directions. Dividing the matched cost and deformation inequalities and taking infima gives the result, including the empty-witness convention and infinite costs. □

B.5 Composition of Audited Comparisons

Lemma B.8 (Closure under Composed Sector Comparisons)**.** *Two consecutive transports satisfying Theorem B.7 compose to a transport of the same kind. If both also intertwine the specified readouts exactly, their composition does so.*

Proof. The ambient and intervention bijections compose and the upper and lower constants multiply. For matched readouts F_1, F_2, F_3 and transports A_1, A_2, the equalities $F_2 A_1 = F_1$ and $F_3 A_2 = F_2$ give $F_3 A_2 A_1 = F_1$. □

A morphism of projective systems gives a map on compatible histories. The finite-budget and resistance obstructions depend respectively on the stated response family and on the matched intervention sector. Inclusions that enlarge the set of histories require their own comparison data.

B.6 Rigidity Hierarchy

Definition B.9 (Rigidity Degree)**.** For a projective system $(S_t, \pi_{t+1\to t})$, define the rigidity degree $\rho \in [0,\infty]$ as:

$$\rho := \sup\left\{ r \ge 0 \;\middle|\; \forall\{\epsilon_t\} : \|\{\epsilon_t\}\|_w < r \implies d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0 \right\}. \tag{41}$$

Theorem B.10 (Stratification Theorem)**.** *The class of observable channels admits a stratification by rigidity degree:*

$$\mathcal{U} = \mathcal{U}_0 \sqcup \mathcal{U}_{(0,\infty)} \sqcup \mathcal{U}_\infty, \tag{42}$$

where:

- $\mathcal{U}_0$*: channels with $\rho = 0$ (zero rigidity, fully deformable);*
- $\mathcal{U}_{(0,\infty)}$*: channels with $0 < \rho < \infty$ (partial rigidity);*
- $\mathcal{U}_\infty$*: channels with $\rho = \infty$ (CCR rigidity).*

Proof. The rigidity degree ρ is well-defined for a specified intervention sector and a common metric by the extended-real supremum in its definition. The stratification follows from the trichotomy of ρ values. □

B.7 Spectral Analysis of Deformations

Definition B.11 (Deformation Spectrum)**.** For a perturbation family $\{\epsilon_t\}$, define the deformation spectrum as the sequence:

$$\sigma(\{\epsilon_t\}) := (\|\epsilon_0\|_{\text{op}}, \|\epsilon_1\|_{\text{op}}, \|\epsilon_2\|_{\text{op}}, \ldots)\,. \tag{43}$$

Proposition B.12 (Spectral Decay Condition)**.** *Let $\{\epsilon_t\}$ have spectrum σ. If there exists $\alpha > 1$ and $C > 0$ such that:*

$$\|\epsilon_t\|_{op} \leq C \cdot t^{-\alpha} \quad \forall t \geq 1, \tag{44}$$

then $\|\{\epsilon_t\}\|_w < \infty$ for any polynomially bounded weight sequence.

Proof. Direct from convergence of $\sum t^{-\alpha}$ for $\alpha > 1$. □

B.8 Completeness Criteria

Definition B.13 (Projective Completeness)**.** A projective system $(S_t, \pi_{t+1\to t})$ is *projectively complete* if for every Cauchy sequence $(x^{(n)})_{n\in\mathbb{N}}$ in $(\mathcal{I}, d_w)$, the limit exists in $\mathcal{I}$.

Theorem B.14 (Completeness Inheritance)**.** *If the coordinate spaces are complete and the bonding maps continuous, every finite-distance component of $(\mathcal{I}, d_w)$ is complete. Under the full-product finiteness condition of Proposition 5.8, this gives completeness on all of $\mathcal{I}$.*

Proof. Apply Proposition 5.8 to the component of any fixed history. Coordinate limits of a Cauchy sequence respect the closed compatibility equations, and the same uniform Cauchy bound yields weighted convergence. □

B.9 Categorical Universality

Proposition B.15 (Universal Property of Identity)**.** *The identity $\mathcal{I} = \varprojlim S_t$ satisfies the universal property: for any object J with compatible maps $\phi_t : J \to S_t$, there exists a unique morphism $\psi : J \to \mathcal{I}$ making all diagrams commute.*

This confirms that rigid identity is not merely a construction but the *terminal object* in the category of compatible families over the projective system [4].

B.10 Relation to Causal Inference

The causal closure property established in Appendix A has implications for causal inference frameworks [5]:

Proposition B.16 (Causal Interventional Invariance)**.** *If $\mathcal{I}$ is a CCR identity, then for any admissible intervention $do(X = x)$ with finite scope:*

$$\mathcal{I}_{\textit{post-intervention}} \cong \mathcal{I}_{\textit{pre-intervention}}. \tag{45}$$

Proof. Finite interventions induce finite-energy perturbations. By CCR rigidity, $M_\Omega = +\infty$ implies no deformation. □

B.11 Computational Complexity Bounds

Theorem B.17 (Complexity Lower Bound for CCR Simulation)**.** *Suppose a deterministic code of B bits must recover any requested prefix of a binary history through horizon t. Exact prefix recovery requires $B \geq t$. For complete decoded histories in the prefix metric $d(x, y) = 2^{-m}$ at first differing position $m \geq 1$, the optimal uniform error is $2^{-(B+1)}$.*

Proof. All 2^t prefixes occur, so distinct prefixes must have distinct codes. For the error bound use 2^{B+1} histories with distinct $(B+1)$-bit prefixes. A shared code cannot decode to the correct $(B+1)$-bit prefix of both, and failure on this prefix gives error at least $2^{-(B+1)}$. Keeping B bits and padding with zeros attains that error. □

Corollary B.18 (Impossibility of Perfect Simulation)**.** *A fixed finite binary budget cannot recover every prefix on the binary-prefix target.*

By contrast, the exact scalar $\sum_{j\geq 1} 2b_j 3^{-j}$ distinguishes all histories, because its leading difference exceeds the remaining tail. On its image the corresponding inverse decoder recovers each finite observation. This model separates numerical precision from real dimension.

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